

## Supplementary tables

Device time and error statistics of every configuration timed or measured on each collection, from the records in `results/v2r` of the study repository. The column *Rings* gives the error in four concentric regions, the quarters of the half-width from the scene center outward. Configurations are labeled as in the paper.

Table 1: Every configuration timed or measured, Panama Canal: device time in seconds, with the pipelined seconds per image in parentheses where a loop was run; error and largest pixel difference (dB), 5 by 5 coherence statistics, amplitude and phase statistics over the brighter half of the pixels, and the error in four rings by distance from the scene center (quarters of the half-width, dB). A dash marks a configuration that was timed without saving its image.

| Device  | Configuration                                                         | Time          | Error | Max   | Coh mean | Coh p0.1 | Amp p99 | Amp max | Ph p99 | Ph max | Rings                 |
|---------|-----------------------------------------------------------------------|---------------|-------|-------|----------|----------|---------|---------|--------|--------|-----------------------|
| L4      | Exact BP, float16 profiles                                            | 103.8 (99.3)  | -56.8 | -62.2 | 1.0000   | 1.000    | 0.922   | 0.03    | 2.5    | 0.2    | 15 -57, -56, -57, -57 |
| L4      | Exact BP, float16 profiles, 16x oversampled                           | 203.3         | -74.5 | -85.2 | 1.0000   | 1.000    | 1.000   | 0.01    | 0.1    | 0.0    | 1 -74, -75, -75, -74  |
| L4      | Exact BP, float32 profiles                                            | 216.4 (216.4) | -56.9 | -62.2 | 1.0000   | 1.000    | 0.923   | 0.03    | 2.5    | 0.2    | 15 -57, -56, -57, -57 |
| L4      | Exact BP, float32 profiles, 16x oversampled                           | 295.2         | -78.5 | -85.5 | 1.0000   | 1.000    | 1.000   | 0.00    | 0.1    | 0.0    | 0 -78, -79, -79, -78  |
| L4      | Factorized BP, float16 products                                       | 13.0          | -56.1 | -64.9 | 1.0000   | 1.000    | 0.843   | 0.04    | 3.2    | 0.3    | 20 -56, -56, -56, -56 |
| L4      | Factorized BP, float16 products, conv. filters                        | 10.9 (12.0)   | -56.2 | -64.2 | 1.0000   | 1.000    | 0.843   | 0.04    | 3.2    | 0.3    | 20 -56, -56, -56, -56 |
| L4      | Factorized BP, float16 throughout, CUDA kernels                       | 2.9 (3.1)     | -58.6 | -64.4 | 1.0000   | 1.000    | 0.886   | 0.03    | 3.2    | 0.2    | 18 -58, -58, -59, -59 |
| L4      | Factorized BP, float16 products, direct ramps                         | 12.8 (13.8)   | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| L4      | Factorized BP, float32, float16 final stage, CUDA kernels             | 3.2 (3.4)     | -59.3 | -64.4 | 1.0000   | 1.000    | 0.886   | 0.03    | 3.2    | 0.2    | 18 -59, -59, -59, -60 |
| L4      | Factorized BP, float32                                                | 29.4          | -57.3 | -64.3 | 1.0000   | 1.000    | 0.886   | 0.03    | 3.4    | 0.2    | 18 -57, -57, -57, -58 |
| L4      | Factorized BP, float32, conv. filters                                 | 13.9 (14.8)   | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| L4      | Factorized BP, float32, CUDA kernels                                  | 3.6 (3.8)     | -59.5 | -64.7 | 1.0000   | 1.000    | 0.886   | 0.03    | 3.2    | 0.2    | 18 -59, -59, -59, -60 |
| L4      | Factorized BP, TF32 products                                          | 19.3          | -56.8 | -65.0 | 1.0000   | 1.000    | 0.842   | 0.04    | 3.2    | 0.2    | 20 -57, -56, -57, -57 |
| L4      | Factorized BP, TF32 products, conv. filters                           | 13.8 (15.0)   | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| L4      | Polar format, float32                                                 | 3.0 (3.2)     | -32.1 | -40.1 | 0.9994   | 0.986    | 0.680   | 0.69    | 20.7   | 6.1    | 70 -40, -37, -35, -30 |
| L4      | Polar format, float32, gather                                         | 2.1 (2.1)     | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v5e | Factorized BP, single-pass, bfloat16 storage                          | 12.7          | -45.8 | -54.2 | 0.9999   | 0.998    | 0.707   | 0.16    | 9.3    | 1.0    | 81 -45, -45, -46, -46 |
| TPU v5e | Factorized BP, six-pass products                                      | 17.6          | -57.6 | -64.5 | 1.0000   | 1.000    | 0.886   | 0.03    | 3.4    | 0.2    | 18 -58, -57, -58, -58 |
| TPU v5e | Factorized BP, six-pass products, direct ramps                        | 13.3          | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v5e | Factorized BP, single-pass products                                   | 8.0           | -45.8 | -54.2 | 0.9999   | 0.998    | 0.707   | 0.16    | 9.3    | 1.0    | 81 -45, -45, -46, -46 |
| TPU v5e | Factorized BP, single-pass products, conv. filters                    | 31.9          | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v5e | Factorized BP, single-pass products, direct ramps                     | 6.5 (6.2)     | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v5e | Factorized BP, single-pass products, fused kernels                    | 2.8 (3.0)     | -45.8 | -54.2 | 0.9999   | 0.998    | 0.705   | 0.16    | 9.4    | 1.0    | 83 -45, -45, -46, -46 |
| TPU v5e | Factorized BP, single-pass products, fused kernels, direct-trig final | 4.5           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v5e | Factorized BP, single-pass products, fused kernels                    | 2.9           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v5e | Factorized BP, single-pass products, fused kernels                    | 3.2           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v5e | Factorized BP, single-pass products, fused kernels, XLA final stage   | 4.4           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v5e | Factorized BP, three-pass products                                    | 13.8          | -57.6 | -64.3 | 1.0000   | 1.000    | 0.884   | 0.03    | 3.3    | 0.2    | 18 -58, -57, -58, -58 |
| TPU v5e | Factorized BP, three-pass products, direct ramps                      | 9.3 (9.0)     | -57.6 | -64.5 | 1.0000   | 1.000    | 0.884   | 0.03    | 3.3    | 0.2    | 18 -58, -57, -58, -58 |
| TPU v5e | Factorized BP, three-pass products, fused kernels                     | 4.6 (4.8)     | -57.6 | -64.5 | 1.0000   | 1.000    | 0.886   | 0.03    | 3.4    | 0.2    | 18 -58, -57, -58, -58 |
| TPU v5e | Factorized BP, three-pass products, fused kernels, direct-trig final  | 6.2           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v5e | Factorized BP, three-pass products, fused kernels                     | 4.7           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v5e | Factorized BP, three-pass products, fused kernels                     | 5.1           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v5e | Factorized BP, three-pass products, fused kernels, XLA final stage    | 5.9           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v5e | Polar format, float32, gather                                         | 97.4          | -32.1 | -40.1 | 0.9994   | 0.986    | 0.680   | 0.69    | 20.7   | 6.1    | 70 -40, -37, -35, -30 |
| TPU v6e | Factorized BP, single-pass, bfloat16 storage                          | 5.2           | -45.8 | -54.1 | 0.9999   | 0.998    | 0.707   | 0.16    | 9.9    | 1.0    | 86 -45, -45, -46, -46 |
| TPU v6e | Factorized BP, six-pass products                                      | 7.2           | -57.5 | -64.2 | 1.0000   | 1.000    | 0.886   | 0.03    | 3.6    | 0.2    | 18 -58, -57, -57, -58 |
| TPU v6e | Factorized BP, six-pass products, direct ramps                        | 5.9           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v6e | Factorized BP, single-pass products                                   | 4.5           | -45.8 | -54.1 | 0.9999   | 0.998    | 0.707   | 0.16    | 9.9    | 1.0    | 86 -45, -45, -46, -46 |
| TPU v6e | Factorized BP, single-pass products, conv. filters                    | 18.6          | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v6e | Factorized BP, single-pass products, direct ramps                     | 3.8 (3.5)     | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v6e | Factorized BP, single-pass products, fused kernels                    | 1.7 (1.8)     | -45.8 | -54.1 | 0.9999   | 0.998    | 0.705   | 0.16    | 10.1   | 1.0    | 86 -45, -45, -46, -46 |
| TPU v6e | Factorized BP, single-pass products, fused kernels, direct-trig final | 2.7           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v6e | Factorized BP, single-pass products, fused kernels                    | 1.7           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v6e | Factorized BP, single-pass products, fused kernels                    | 1.9           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v6e | Factorized BP, single-pass products, fused kernels, XLA final stage   | 2.8           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v6e | Factorized BP, three-pass products                                    | 6.1           | -57.5 | -64.0 | 1.0000   | 1.000    | 0.884   | 0.03    | 3.6    | 0.2    | 18 -58, -57, -57, -58 |
| TPU v6e | Factorized BP, three-pass products, direct ramps                      | 4.6 (4.3)     | -57.5 | -64.2 | 1.0000   | 1.000    | 0.884   | 0.03    | 3.6    | 0.2    | 18 -58, -57, -57, -58 |
| TPU v6e | Factorized BP, three-pass products, fused kernels                     | 2.4 (2.5)     | -57.5 | -64.2 | 1.0000   | 1.000    | 0.886   | 0.03    | 3.6    | 0.2    | 18 -58, -57, -57, -58 |
| TPU v6e | Factorized BP, three-pass products, fused kernels, direct-trig final  | 3.3           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v6e | Factorized BP, three-pass products, fused kernels                     | 2.3           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v6e | Factorized BP, three-pass products, fused kernels                     | 2.6           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v6e | Factorized BP, three-pass products, fused kernels, XLA final stage    | 3.4           | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| TPU v6e | Polar format, float32                                                 | 105.3 (107.5) | -32.1 | -40.1 | 0.9994   | 0.986    | 0.680   | 0.69    | 20.7   | 6.1    | 70 -40, -37, -35, -30 |
| TPU v6e | Polar format, float32, gather                                         | 105.4         | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| CPU     | Factorized BP, float32, conv. filters, direct ramps                   | 217.4         | -58.0 | -64.4 | 1.0000   | 1.000    | 0.886   | 0.03    | 3.2    | 0.2    | 18 -58, -58, -58, -58 |
| CPU     | Factorized BP, float32, C++ kernels                                   | 12.4 (12.1)   | -59.6 | -64.6 | 1.0000   | 1.000    | 0.886   | 0.03    | 3.2    | 0.2    | 18 -59, -59, -59, -60 |
| CPU     | Factorized BP, float64, conv. filters, direct ramps                   | 616.3         | -59.6 | -64.6 | 1.0000   | 1.000    | 0.886   | 0.03    | 3.2    | 0.2    | 18 -59, -59, -59, -60 |
| CPU     | Polar format, float32                                                 | 17.0          | -     | -     | -        | -        | -       | -       | -      | -      | -                     |
| CPU     | Polar format, float32, gather                                         | 11.3          | -32.1 | -40.2 | 0.9994   | 0.986    | 0.679   | 0.69    | 20.1   | 6.1    | 70 -40, -37, -35, -30 |

Table 2: Every configuration timed or measured, Melbourne: device time in seconds, with the pipelined seconds per image in parentheses where a loop was run; error and largest pixel difference (dB), 5 by 5 coherence statistics, amplitude and phase statistics over the brighter half of the pixels, and the error in four rings by distance from the scene center (quarters of the half-width, dB). A dash marks a configuration that was timed without saving its image.

| Device  | Configuration                                             | Time          | Error | Max   | Coh mean | Coh p0.1 | Coh min | Amp p99 | Amp max | Ph p99 | Ph max | Rings              |
|---------|-----------------------------------------------------------|---------------|-------|-------|----------|----------|---------|---------|---------|--------|--------|--------------------|
| L4      | Exact BP, float16 profiles                                | 102.7 (101.8) | -56.8 | -61.3 | 1.0000   | 1.000    | 0.957   | 0.04    | 8.2     | 0.3    | 22     | -57, -57, -57      |
| L4      | Exact BP, float32 profiles                                | 248.2 (247.4) | -56.8 | -61.3 | 1.0000   | 1.000    | 0.958   | 0.04    | 8.2     | 0.3    | 22     | -57, -57, -57      |
| L4      | Factorized BP, float16 products                           | 14.8          | -56.5 | -57.5 | 1.0000   | 1.000    | 0.929   | 0.05    | 2.8     | 0.3    | 24     | -57, -56, -57      |
| L4      | Factorized BP, float16 products, conv. filters            | 12.1 (13.1)   | -56.5 | -57.5 | 1.0000   | 1.000    | 0.929   | 0.05    | 2.8     | 0.3    | 24     | -57, -56, -57      |
| L4      | Factorized BP, float16 throughout, CUDA kernels           | 2.9 (3.2)     | -58.2 | -58.5 | 1.0000   | 1.000    | 0.947   | 0.04    | 2.4     | 0.3    | 20     | -59, -58, -58      |
| L4      | Factorized BP, float16 products, direct ramps             | 14.2          | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| L4      | Factorized BP, float32, float16 final stage, CUDA kernels | 3.1 (3.5)     | -58.9 | -58.5 | 1.0000   | 1.000    | 0.947   | 0.04    | 2.4     | 0.2    | 21     | -59, -58, -59      |
| L4      | Factorized BP, float32                                    | 33.4          | -58.0 | -57.4 | 1.0000   | 1.000    | 0.947   | 0.04    | 2.4     | 0.3    | 21     | -58, -58, -58      |
| L4      | Factorized BP, float32, conv. filters                     | 15.4 (16.3)   | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| L4      | Factorized BP, float32, CUDA kernels                      | 3.6 (4.0)     | -59.1 | -58.2 | 1.0000   | 1.000    | 0.947   | 0.04    | 2.4     | 0.2    | 21     | -59, -59, -59      |
| L4      | Factorized BP, TF32 products                              | 22.4          | -57.2 | -57.4 | 1.0000   | 1.000    | 0.928   | 0.05    | 2.8     | 0.3    | 24     | -57, -57, -57      |
| L4      | Factorized BP, TF32 products, conv. filters               | 15.1 (15.9)   | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| L4      | Polar format, float32                                     | 3.2 (3.3)     | -33.4 | -46.3 | 0.9997   | 0.994    | 0.873   | 0.52    | 23.9    | 4.1    | 180    | -42, -39, -36, -30 |
| L4      | Polar format, float32, gather                             | 2.2 (2.3)     | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v5e | Factorized BP, single-pass, bfloat16 storage              | 10.7          | -46.8 | -54.5 | 1.0000   | 0.999    | 0.858   | 0.17    | 12.9    | 1.1    | 172    | -48, -46, -47, -47 |
| TPU v5e | Factorized BP, six-pass products                          | 17.6          | -57.7 | -57.3 | 1.0000   | 1.000    | 0.947   | 0.04    | 2.4     | 0.3    | 21     | -58, -57, -58, -58 |
| TPU v5e | Factorized BP, single-pass products, direct ramps         | 14.7          | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v5e | Factorized BP, single-pass products, conv. filters        | 7.9           | -46.8 | -54.5 | 1.0000   | 0.999    | 0.858   | 0.17    | 12.9    | 1.1    | 172    | -48, -46, -47, -47 |
| TPU v5e | Factorized BP, single-pass products, direct ramps         | 32.2          | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v5e | Factorized BP, single-pass products, conv. filters        | 6.8 (6.5)     | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v5e | Factorized BP, single-pass products, direct ramps         | 2.7 (2.9)     | -46.8 | -54.5 | 1.0000   | 0.999    | 0.859   | 0.17    | 13.1    | 1.1    | 171    | -48, -46, -47, -47 |
| TPU v5e | Factorized BP, three-pass products                        | 13.1          | -57.7 | -57.6 | 1.0000   | 1.000    | 0.946   | 0.04    | 2.5     | 0.3    | 21     | -58, -57, -58, -58 |
| TPU v5e | Factorized BP, three-pass products, fused kernels         | 10.0 (9.7)    | -57.7 | -57.3 | 1.0000   | 1.000    | 0.946   | 0.04    | 2.5     | 0.3    | 21     | -58, -57, -58, -58 |
| TPU v5e | Factorized BP, three-pass products, direct ramps          | 4.7 (5.0)     | -57.7 | -57.3 | 1.0000   | 1.000    | 0.947   | 0.04    | 2.4     | 0.3    | 21     | -58, -57, -58, -58 |
| TPU v5e | Factorized BP, three-pass products, fused kernels         | 62.5          | -33.4 | -46.3 | 0.9997   | 0.994    | 0.873   | 0.52    | 23.9    | 4.1    | 180    | -42, -39, -36, -30 |
| TPU v5e | Polar format, float32, gather                             | 6.3           | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v6e | Factorized BP, single-pass, bfloat16 storage              | 5.6           | -46.8 | -53.7 | 1.0000   | 0.999    | 0.859   | 0.17    | 13.1    | 1.1    | 172    | -47, -46, -47, -47 |
| TPU v6e | Factorized BP, six-pass products                          | 7.5           | -57.5 | -55.9 | 1.0000   | 1.000    | 0.947   | 0.04    | 2.4     | 0.3    | 21     | -57, -57, -58, -58 |
| TPU v6e | Factorized BP, single-pass products                       | 6.3           | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v6e | Factorized BP, single-pass products, conv. filters        | 4.4           | -46.8 | -53.7 | 1.0000   | 0.999    | 0.859   | 0.17    | 13.2    | 1.1    | 172    | -47, -46, -47, -47 |
| TPU v6e | Factorized BP, single-pass products, direct ramps         | 18.0          | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v6e | Factorized BP, single-pass products, fused kernels        | 3.9 (3.6)     | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v6e | Factorized BP, single-pass products, fused kernels        | 1.6 (1.7)     | -46.8 | -53.8 | 1.0000   | 0.999    | 0.857   | 0.17    | 13.2    | 1.1    | 172    | -48, -46, -47, -47 |
| TPU v6e | Factorized BP, three-pass products                        | 6.0           | -57.4 | -56.1 | 1.0000   | 1.000    | 0.946   | 0.04    | 2.5     | 0.3    | 21     | -57, -57, -58, -58 |
| TPU v6e | Factorized BP, three-pass products, direct ramps          | 4.6 (4.4)     | -57.5 | -55.9 | 1.0000   | 1.000    | 0.946   | 0.04    | 2.5     | 0.3    | 21     | -57, -57, -58, -58 |
| TPU v6e | Factorized BP, three-pass products, fused kernels         | 2.1 (2.3)     | -57.5 | -56.0 | 1.0000   | 1.000    | 0.947   | 0.04    | 2.4     | 0.3    | 21     | -57, -57, -58, -58 |
| TPU v6e | Polar format, float32                                     | 101.7 (104.0) | -33.4 | -46.3 | 0.9997   | 0.994    | 0.873   | 0.52    | 23.9    | 4.1    | 180    | -42, -39, -36, -30 |
| TPU v6e | Polar format, float32, gather                             | 101.8         | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| CPU     | Factorized BP, float32, conv. filters, direct ramps       | 234.7         | -58.0 | -56.4 | 1.0000   | 1.000    | 0.947   | 0.04    | 2.4     | 0.3    | 21     | -58, -58, -58, -58 |
| CPU     | Factorized BP, float32, C++ kernels                       | 12.5 (12.3)   | -59.1 | -58.3 | 1.0000   | 1.000    | 0.947   | 0.04    | 2.4     | 0.2    | 21     | -59, -59, -59, -59 |
| CPU     | Factorized BP, float64, conv. filters, direct ramps       | 642.0         | -59.1 | -58.3 | 1.0000   | 1.000    | 0.947   | 0.04    | 2.4     | 0.2    | 21     | -59, -59, -59, -59 |
| CPU     | Polar format, float32                                     | 17.8          | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| CPU     | Polar format, float32, gather                             | 11.6          | -33.5 | -46.7 | 0.9997   | 0.994    | 0.873   | 0.52    | 23.8    | 4.1    | 179    | -42, -39, -36, -30 |

Table 3: Every configuration timed or measured, Iowa farmland: device time in seconds, with the pipelined seconds per image in parentheses where a loop was run; error and largest pixel difference (dB), 5 by 5 coherence statistics, amplitude and phase statistics over the brighter half of the pixels, and the error in four rings by distance from the scene center (quarters of the half-width, dB). A dash marks a configuration that was timed without saving its image.

| Device  | Configuration                                             | Time          | Error | Max   | Coh mean | Coh p0.1 | Coh min | Amp p99 | Amp max | Ph p99 | Ph max | Rings              |
|---------|-----------------------------------------------------------|---------------|-------|-------|----------|----------|---------|---------|---------|--------|--------|--------------------|
| L4      | Exact BP, float16 profiles                                | 63.8 (63.4)   | -60.4 | -63.3 | 1.0000   | 1.000    | 0.994   | 0.02    | 1.3     | 0.1    | 8      | -60, -60, -61, -60 |
| L4      | Exact BP, float32 profiles                                | 142.3 (141.8) | -60.5 | -63.3 | 1.0000   | 1.000    | 0.994   | 0.02    | 1.2     | 0.1    | 8      | -60, -60, -61, -60 |
| L4      | Factorized BP, float16 products                           | 10.6          | -57.2 | -63.2 | 1.0000   | 1.000    | 0.974   | 0.03    | 1.4     | 0.2    | 9      | -57, -57, -57, -57 |
| L4      | Factorized BP, float16 products, conv. filters            | 8.9 (9.6)     | -57.2 | -63.0 | 1.0000   | 1.000    | 0.974   | 0.03    | 1.4     | 0.2    | 9      | -57, -57, -57, -57 |
| L4      | Factorized BP, float16 throughput, CUDA kernels           | 1.8 (2.0)     | -59.0 | -62.9 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.1     | 0.1    | 7      | -59, -59, -59, -59 |
| L4      | Factorized BP, float16 products, direct ramps             | 10.5          | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| L4      | Factorized BP, float32, float16 final stage, CUDA kernels | 2.0 (2.2)     | -59.8 | -62.9 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.1     | 0.1    | 7      | -60, -60, -60, -60 |
| L4      | Factorized BP, float32                                    | 22.0          | -58.9 | -63.0 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.2     | 0.1    | 7      | -59, -59, -59, -59 |
| L4      | Factorized BP, float32, conv. filters                     | 10.3 (10.9)   | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| L4      | Factorized BP, float32, CUDA kernels                      | 2.3 (2.5)     | -60.0 | -63.0 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.1     | 0.1    | 7      | -60, -60, -60, -60 |
| L4      | Factorized BP, TF32 products                              | 14.5          | -58.1 | -63.2 | 1.0000   | 1.000    | 0.974   | 0.03    | 1.4     | 0.2    | 9      | -58, -58, -58, -58 |
| L4      | Factorized BP, TF32 products, conv. filters               | 9.9 (10.5)    | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| L4      | Polar format, float32                                     | 2.1 (2.2)     | -35.3 | -35.3 | 0.9998   | 0.998    | 0.973   | 0.33    | 9.5     | 2.7    | 38     | -40, -39, -37, -33 |
| L4      | Polar format, float32, gather                             | 1.6 (1.7)     | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v5e | Factorized BP, single-pass, bfloat16 storage              | 5.0           | -46.9 | -62.0 | 1.0000   | 1.000    | 0.957   | 0.10    | 1.8     | 0.6    | 30     | -47, -47, -47, -47 |
| TPU v5e | Factorized BP, six-pass products                          | 10.5          | -58.7 | -63.0 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.1     | 0.1    | 8      | -59, -59, -59, -59 |
| TPU v5e | Factorized BP, single-pass products, direct ramps         | 8.6           | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v5e | Factorized BP, single-pass products, conv. filters        | 5.4           | -46.9 | -62.0 | 1.0000   | 1.000    | 0.957   | 0.10    | 1.8     | 0.6    | 30     | -47, -47, -47, -47 |
| TPU v5e | Factorized BP, single-pass products, direct ramps         | 10.9          | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v5e | Factorized BP, single-pass products, fused kernels        | 4.4 (4.2)     | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v5e | Factorized BP, single-pass products, direct ramps         | 1.8 (2.0)     | -46.9 | -61.9 | 1.0000   | 1.000    | 0.956   | 0.10    | 1.8     | 0.6    | 30     | -47, -47, -47, -47 |
| TPU v5e | Factorized BP, three-pass products                        | 7.9           | -58.7 | -62.7 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.1     | 0.1    | 8      | -59, -58, -59, -59 |
| TPU v5e | Factorized BP, three-pass products, direct ramps          | 6.1 (5.9)     | -58.7 | -63.0 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.1     | 0.1    | 8      | -59, -58, -59, -59 |
| TPU v5e | Factorized BP, three-pass products, fused kernels         | 3.0 (3.2)     | -58.7 | -62.9 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.1     | 0.1    | 8      | -59, -59, -59, -59 |
| TPU v5e | Polar format, float32, gather                             | 44.2          | -35.3 | -35.3 | 0.9998   | 0.998    | 0.973   | 0.33    | 9.5     | 2.7    | 38     | -40, -39, -37, -33 |
| TPU v6e | Factorized BP, single-pass, bfloat16 storage              | 3.2           | -46.9 | -62.0 | 1.0000   | 1.000    | 0.957   | 0.10    | 1.8     | 0.6    | 29     | -47, -47, -47, -47 |
| TPU v6e | Factorized BP, six-pass products                          | 4.2           | -58.9 | -63.0 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.2     | 0.1    | 8      | -59, -59, -59, -59 |
| TPU v6e | Factorized BP, single-pass products                       | 4.2           | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v6e | Factorized BP, single-pass products, conv. filters        | 2.9           | -46.9 | -62.0 | 1.0000   | 1.000    | 0.957   | 0.10    | 1.8     | 0.6    | 29     | -47, -47, -47, -47 |
| TPU v6e | Factorized BP, single-pass products, direct ramps         | 9.5           | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v6e | Factorized BP, single-pass products, fused kernels        | 3.1 (2.9)     | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v6e | Factorized BP, single-pass products, direct ramps         | 1.1 (1.2)     | -46.9 | -61.9 | 1.0000   | 1.000    | 0.956   | 0.10    | 1.8     | 0.6    | 30     | -47, -47, -47, -47 |
| TPU v6e | Factorized BP, single-pass products, fused kernels        | 3.5           | -58.8 | -62.7 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.2     | 0.1    | 8      | -59, -59, -59, -59 |
| TPU v6e | Factorized BP, three-pass products                        | 3.5 (3.3)     | -58.8 | -63.0 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.2     | 0.1    | 8      | -59, -59, -59, -59 |
| TPU v6e | Factorized BP, three-pass products, direct ramps          | 1.6 (1.7)     | -58.9 | -63.0 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.2     | 0.1    | 8      | -59, -59, -59, -59 |
| TPU v6e | Factorized BP, three-pass products, fused kernels         | 57.8 (59.3)   | -35.3 | -35.3 | 0.9998   | 0.998    | 0.973   | 0.33    | 9.5     | 2.7    | 38     | -40, -39, -37, -33 |
| TPU v6e | Polar format, float32                                     | 57.8          | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| TPU v6e | Polar format, float32, gather                             | 191.0         | -59.2 | -62.9 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.2     | 0.1    | 7      | -59, -59, -59, -59 |
| CPU     | Factorized BP, float32, conv. filters, direct ramps       | 8.5 (8.5)     | -60.1 | -63.0 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.1     | 0.1    | 7      | -61, -60, -60, -60 |
| CPU     | Factorized BP, float32, C++ kernels                       | 409.9         | -60.1 | -63.0 | 1.0000   | 1.000    | 0.981   | 0.02    | 1.1     | 0.1    | 7      | -61, -60, -60, -60 |
| CPU     | Factorized BP, float64, conv. filters, direct ramps       | 12.1          | -     | -     | -        | -        | -       | -       | -       | -      | -      | -                  |
| CPU     | Polar format, float32                                     | 8.2           | -35.4 | -35.4 | 0.9998   | 0.998    | 0.973   | 0.33    | 9.5     | 2.7    | 38     | -40, -39, -37, -33 |